Sequences & Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 9$ and common difference $d = 6$, find a_{10}.

2 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 15$ and $d = -4$, find a_{11}.

3 [Foundational] • Target: Arithmetic series

Find S_{10} for the arithmetic sequence with $a_1 = 9$, $d = 6$ (using $a_{10} = 63$ from above).

4 [Foundational] • Target: Geometric sequences

For the geometric sequence with $a_1 = 6$ and common ratio $r = 2$, find a_8.

5 [Foundational] • Target: Geometric sequences

A geometric sequence has $a_1 = 2$ and common ratio $r = 5$. Find the smallest n for which $a_n = 1250$.

6 [Foundational] • Target: Geometric series (finite)

Find S_8 for the geometric sequence with $a_1 = 6$, $r = 2$ (using $a_8 = 768$ from above).

7 [Foundational] • Target: Infinite geometric series

Find the sum of the infinite geometric series $36 + 12 + 4 + \cdots$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Infinite geometric series

Find the sum of the infinite geometric series $12 - 4 + \frac{4}{3} - \cdots$.

9 [Intermediate] • Target: Mathematical Induction

A geometric sequence has its first term a_1 and a common ratio r. The sum of the first three terms is 26, and the sum of the first six terms is 728. If $a_1 = 2$, find the value of r.

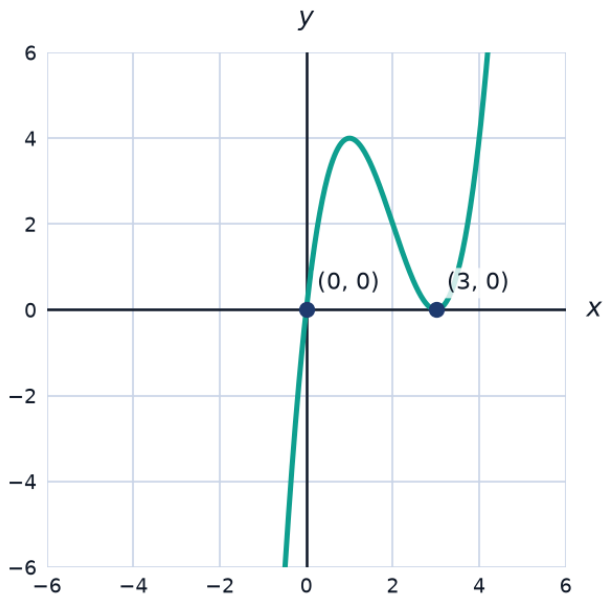
10 [Intermediate] • Target: Arithmetic sequences

A sequence is defined by the recurrence relation $a_{n+1} = 2a_n - 3$ for $n \geq 1$, with $a_1 = 5$. Consider a second sequence, b_n, where $b_n = a_n - k$ for some constant k. If the sequence b_n is a geometric sequence, find the value of k and the common ratio of the sequence b_n.

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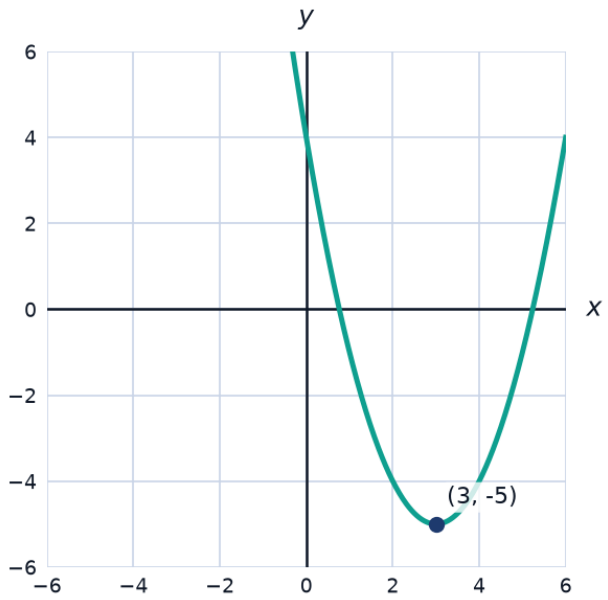
11 **[Intermediate]** • **Target: Arithmetic series**

Consider the function $f(x) = x^3 - 6x^2 + 9x$. Find the area of the region bounded by the graph of $f(x)$ and the x-axis.



12 **[Intermediate]** • **Target: Geometric sequences**

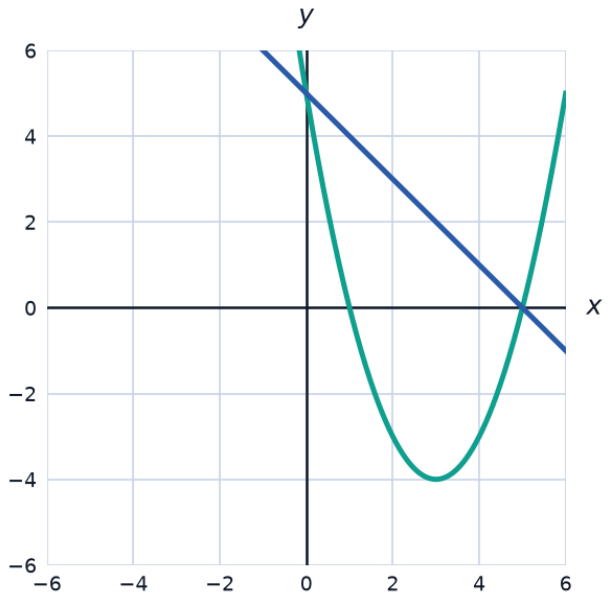
The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph has a vertex at $(3, -5)$. Determine the value of the parameter k.



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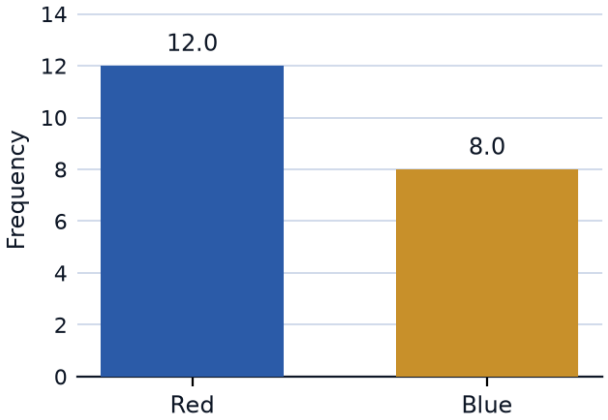
13 [Intermediate] • Target: Geometric series (finite)

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?



14 [Intermediate] • Target: Infinite geometric series

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

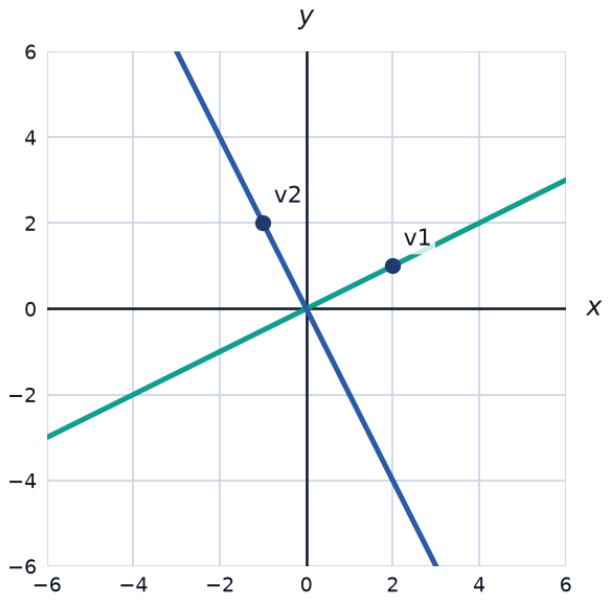


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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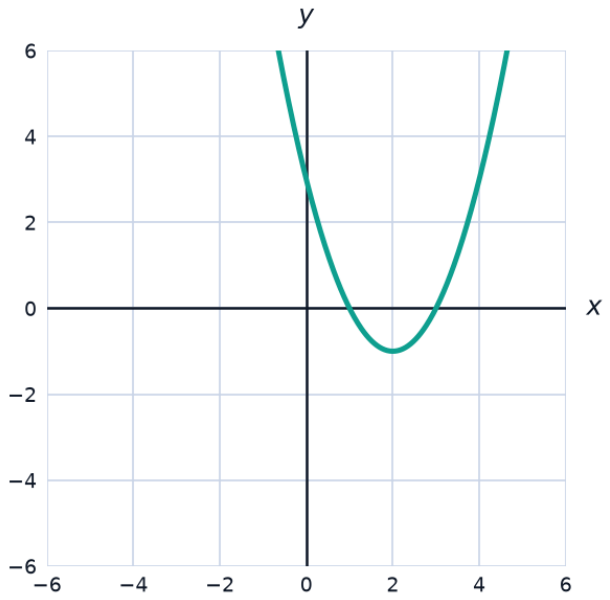
15 [Challenging] • Target: Advanced Arithmetic & Geometric Progressions • Cross-Lesson: 1.01 Complex Numbers

Consider a linear transformation T represented by the matrix $A = (2, 1\ 1, 3)$. Let $P = (x; y)$ be a point in the plane. The transformation maps P to P'. If the area of the region transformed by T from a unit square in the xy-plane is A, and the determinant of A is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is 5.



16 [Challenging] • Target: Summation Notation & Formulas • Cross-Lesson: 3.01 Functions & Graphs

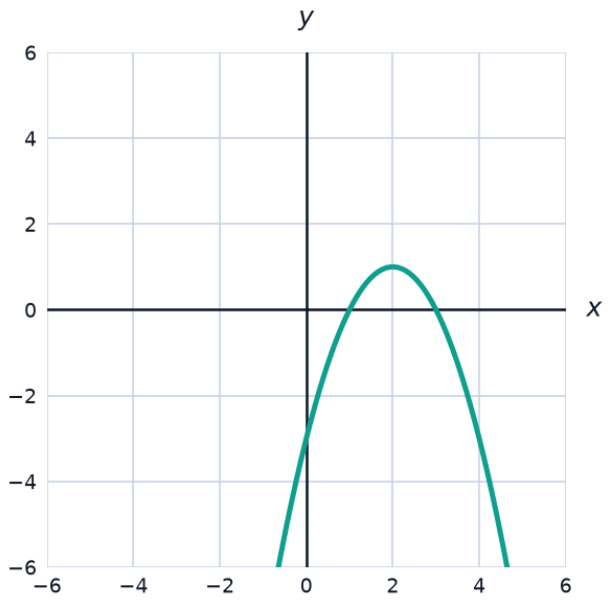
Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?



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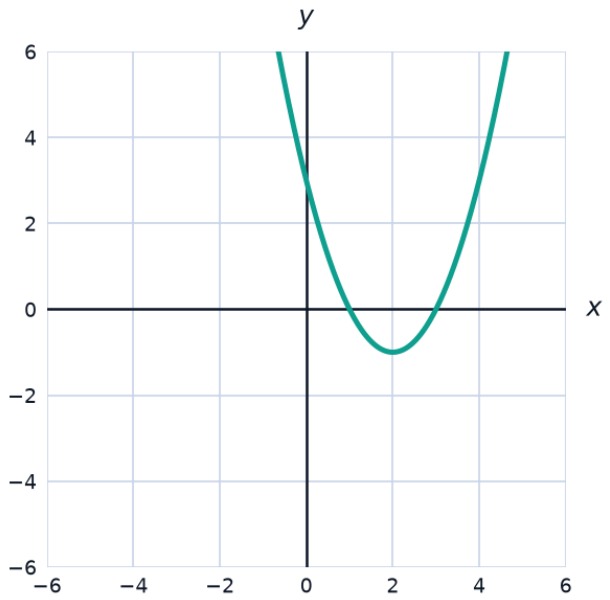
17 [Challenging] • Target: Infinite Geometric Series & Repeating Decimals • Cross-Lesson: 4.01 Matrices & Vectors

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the y-coordinate of the vertex of the parabola.



18 [Challenging] • Target: Mathematical Induction • Cross-Lesson: 5.01 Introductory Calculus

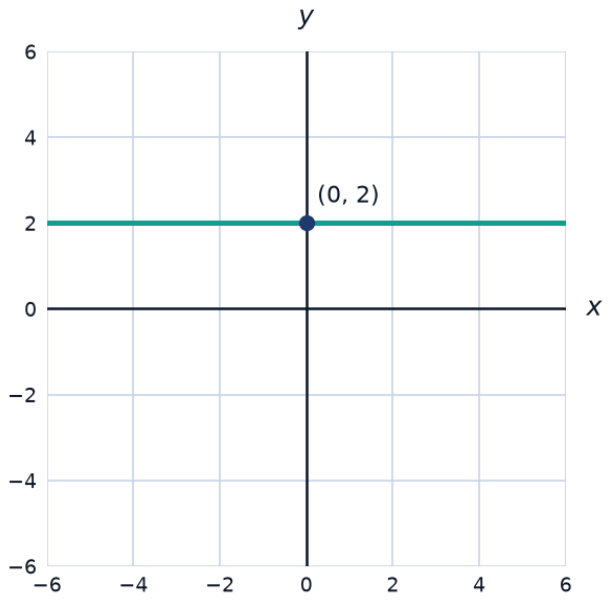
Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. What is the y-intercept of this parabola?



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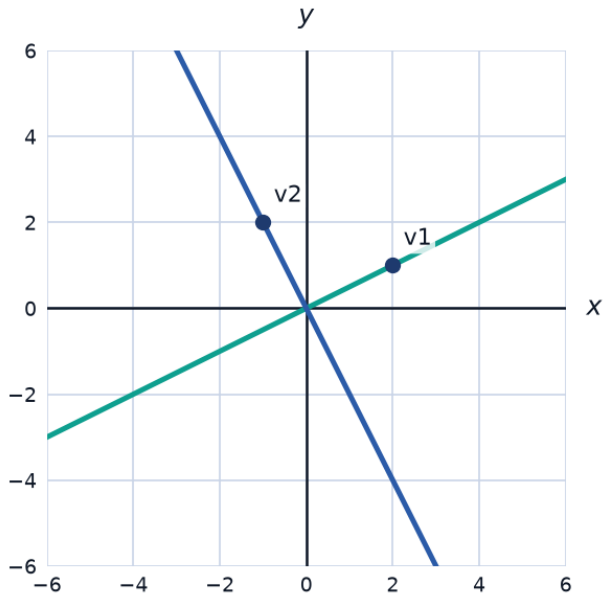
19 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 6.01 Probability

The graph of a linear function $f(x)$ is shown below. What is the value of $f(0)$?



20 [Challenging] • Target: Arithmetic series • Cross-Lesson: 1.01 Complex Numbers

Consider the matrix $A = (2, 0; 0, 3)$. When the matrix A is applied to the vector $\mathbf{v} = (x; y)$, the resulting vector is $A\mathbf{v} = (6; 9)$. What is the original vector $\mathbf{v}$?



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 0$ and common difference $d = 9$, find a_8.

a. 63

b. 65

c. 54

d. 72

22 [Foundational] • Target: Arithmetic series

For the arithmetic sequence with $a_1 = -11$ and common difference $d = -5$, find a_6.

a. -36

b. -34

c. -41

d. -31

23 [Foundational] • Target: Geometric sequences

For the arithmetic sequence with $a_1 = -6$ and common difference $d = 6$, find a_7.

a. 36

b. 30

c. 32

d. 24

24 [Foundational] • Target: Geometric series (finite)

For the arithmetic sequence with $a_1 = 15$ and common difference $d = -9$, find a_7.

a. -30

b. -48

c. -37

d. -39

25 [Foundational] • Target: Infinite geometric series

For the arithmetic sequence with $a_1 = -6$ and common difference $d = -1$, find a_7.

a. -13

b. -11

c. -10

d. -12

26 [Foundational] • Target: Advanced Arithmetic & Geometric Progressions

For the arithmetic sequence with $a_1 = -3$ and common difference $d = 9$, find a_7.

a. 53

b. 42

c. 51

d. 60

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27 [Foundational] • Target: Summation Notation & Formulas

For the arithmetic sequence with $a_1 = 4$ and common difference $d = -3$, find a_{13}.

a. -30

b. -32

c. -35

d. -29

28 [Foundational] • Target: Infinite Geometric Series & Repeating Decimals

Find S_6 for the arithmetic sequence with $a_1 = -13$, $d = -1$.

a. -94

b. -89

c. -97

d. -93

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mathematical Induction

Find S_4 for the arithmetic sequence with $a_1 = 2$, $d = -7$.

a. -30

b. -41

c. -38

d. -34

30 [Intermediate] • Target: Arithmetic sequences

Find S_{10} for the arithmetic sequence with $a_1 = 3$, $d = -8$.

a. -326

b. -330

c. -338

d. -334

31 [Intermediate] • Target: Arithmetic series

Find S_8 for the arithmetic sequence with $a_1 = 1$, $d = -5$.

a. -132

b. -136

c. -128

d. -137

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32 [Intermediate] • Target: Geometric sequences

Find S_{10} for the arithmetic sequence with $a_1 = 20$, $d = -7$.

a. -122

b. -115

c. -119

d. -111

33 [Intermediate] • Target: Geometric series (finite)

Find S_8 for the arithmetic sequence with $a_1 = -15$, $d = -4$.

a. -228

b. -226

c. -236

d. -232

34 [Intermediate] • Target: Infinite geometric series

Find S_{16} for the arithmetic sequence with $a_1 = -12$, $d = 3$.

a. 168

b. 164

c. 171

d. 172

35 [Intermediate] • Target: Advanced Arithmetic & Geometric Progressions

For the geometric sequence with $a_1 = -9$ and common ratio $r = -3$, find a_4.

a. 243

b. 241

c. 245

d. -243

36 [Intermediate] • Target: Summation Notation & Formulas

For the geometric sequence with $a_1 = 9$ and common ratio $r = 3$, find a_3.

a. -81

b. 79

c. 81

d. 83

37 [Intermediate] • Target: Infinite Geometric Series & Repeating Decimals

Find S_4 for the geometric sequence with $a_1 = -6$, $r = 3$.

a. -240

b. -238

c. -246

d. -242

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Mathematical Induction • Cross-Lesson: 5.01 Introductory Calculus

Find S_6 for the geometric sequence with $a_1 = 8$, $r = 3$.

a. 2920

b. 2912

c. 2914

d. 2910

39 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 6.01 Probability

Find the sum of the infinite geometric series $16 - 8 + 4 \cdots$.

a. $\frac{38}{3}$

b. $\frac{32}{3}$

c. $\frac{26}{3}$

d. $-\frac{32}{3}$

40 [Challenging] • Target: Arithmetic series • Cross-Lesson: 1.01 Complex Numbers

Find the sum of the infinite geometric series $144 - 24 + 4 \cdots$.

a. $\frac{878}{7}$

b. $\frac{850}{7}$

c. $\frac{864}{7}$

d. $-\frac{864}{7}$

41 [Challenging] • Target: Advanced Arithmetic & Geometric Progressions • Cross-Lesson: 3.01 Functions & Graphs

The sum of three consecutive terms of an arithmetic progression is 27, and their product is 504. What is the largest of the three terms?

a. 14

b. 16

c. 18

d. 21

42 [Challenging] • Target: Advanced Arithmetic & Geometric Progressions • Cross-Lesson: 4.01 Matrices & Vectors

What is the sum of the series $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \ldots$?

a. 3

b. $\frac{3}{2}$

c. 2

d. $\frac{5}{2}$

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43 [Challenging] • Target: Summation Notation & Formulas • Cross-Lesson: 5.01 Introductory Calculus

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

a. 1

b. $\frac{1}{2}$

c. 2

d. Diverges

44 [Challenging] • Target: Infinite Geometric Series & Repeating Decimals • Cross-Lesson: 6.01 Probability

Express $0.\overline{36}$ as a reduced fraction.

a. $\frac{4}{11}$

b. $\frac{36}{100}$

c. $\frac{1}{3}$

d. $\frac{12}{33}$

45 [Challenging] • Target: Mathematical Induction • Cross-Lesson: 1.01 Complex Numbers

In proving $1 + 2 + \ldots + n = \frac{n(n+1)}{2}$, what is the inductive step?

a. Add $(k + 1)$ to both sides of the hypothesis for $n = k$

b. Multiply both sides by 2

c. Subtract k from n

d. Differentiate with respect to n

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